

| Project Information |                                                |  |          |
|---------------------|------------------------------------------------|--|----------|
| Group ID            | 3                                              |  | Semester |
| 1. Student Name     | Mohamed Hamad                                  |  | 1        |
| 2. Student Name     | Fang Xiao                                      |  | 1        |
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|                     |                                                |  |          |
| Project Title       | Distributed Inventory Management System (DIMS) |  |          |

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| <p><b>Project Description</b></p> | <p>The Distributed Inventory Management System (DIMS) is a distributed system designed to manage shared inventory across multiple scanner clients. The architecture employs a hybrid communication strategy: standard HTTP interfaces for external clients and a custom socket-based middleware for internal coordination.</p> <p><b>1. System Model</b></p> <ul style="list-style-type: none"> <li>• <b>Topology:</b> A cluster of server nodes forming a dynamic logical ring, plus multiple independent scanner clients.</li> <li>• <b>Interaction Model:</b> the system relies on timeouts and connection failures to detect unresponsive nodes.</li> <li>• it will be tolerant to Crashes.</li> </ul> <p><b>2. Hybrid Middleware Architecture</b></p> <p>We implement a layered communication stack to balance interoperability with the requirement of building distributed middleware from scratch:</p> <ul style="list-style-type: none"> <li>• <b>External Layer (Client ↔ System):</b> <ul style="list-style-type: none"> <li>○ <b>HTTP REST:</b> Provides a lightweight, standard interface for scanner clients to submit inventory updates (POST requests).</li> </ul> </li> <li>• <b>Internal Layer (Server ↔ Server):</b> <ul style="list-style-type: none"> <li>○ <b>Custom TCP Protocol:</b> handles core distributed mechanisms (election, replication, heartbeats); establishes persistent TCP connections between logical neighbors.</li> <li>○ <b>UDP (Broadcast/Multicast):</b> exclusively for <b>dynamic discovery</b>, allowing nodes to locate the current leader without prior configuration.</li> </ul> </li> <li>• <b>Message Protocol (Internal):</b> Custom application-level framing and JSON serialization over internal TCP connections.</li> </ul> <p><b>3. Leader Election &amp; Discovery</b></p> <ul style="list-style-type: none"> <li>• <b>Dynamic Host Discovery:</b> <ul style="list-style-type: none"> <li>○ Clients and servers use UDP broadcast to locate the active leader's HTTP endpoint.</li> <li>○ Each server maintains knowledge only of its successor in the logical ring.</li> </ul> </li> <li>• <b>Ring-based Election:</b> <ul style="list-style-type: none"> <li>○ Triggered by internal TCP connection failures or heartbeat timeouts.</li> <li>○ Servers communicate strictly with their logical neighbors in the ring to elect a leader based on the highest Node ID.</li> </ul> </li> </ul> <p><b>4. Replication &amp; Persistence</b></p> <ul style="list-style-type: none"> <li>• <b>Storage:</b> An in-memory key–value store with local persistence via a write-ahead log (WAL), implemented as an append-only JSON file.</li> <li>• <b>Passive Replication:</b> Follower nodes apply received updates forwarded by the leader. <ul style="list-style-type: none"> <li>○ <b>Primary:</b> The elected leader receives HTTP requests and logs updates locally.</li> <li>○ <b>Backup:</b> Updates are propagated to follower nodes through the <b>internal TCP ring</b>.</li> </ul> </li> </ul> <p><b>5. Fault Tolerance</b></p> <ul style="list-style-type: none"> <li>• <b>Failure Detection:</b> <ul style="list-style-type: none"> <li>○ <b>Internal:</b> Broken TCP connections or heartbeat timeouts on the ring trigger ring repair (skipping the unresponsive node) and leader re-election.</li> <li>○ <b>External:</b> Clients detect leader failure through HTTP timeouts (e.g., no response or 503) and re-initiate UDP-based discovery.</li> </ul> </li> <li>• <b>Recovery:</b> A recovering node reloads its local persistent log and rejoins the ring as a follower node.</li> </ul> <p><b>6. Implementation Plan</b></p> <ul style="list-style-type: none"> <li>• <b>Backend:</b> Python <ul style="list-style-type: none"> <li>○ http.server or Flask for handling client HTTP requests</li> <li>○ Socket-based internal middleware for server-to-server communication</li> </ul> </li> <li>• <b>Storage:</b> <ul style="list-style-type: none"> <li>○ Local file-based persistence using an append-only JSON log.</li> </ul> </li> <li>• <b>Middleware Scope:</b> <ul style="list-style-type: none"> <li>○ <b>TCP:</b> Reliable communication for internal replication and leader election logic.</li> <li>○ <b>UDP:</b> Unreliable but lightweight mechanism for dynamic discovery.</li> </ul> </li> </ul> |
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## System Architecture Desin

Include a clear diagram of your system architecture (key components and connections between them).



